

1. Define **Data Science** and explain its advantages in an organization.
2. Explain **applications of Data Science** in different domains.
3. Explain **types of data**:
 - Structured
 - Semi-structured
 - Unstructured
 - Data Streams
4. Compare **Qualitative and Quantitative data** with examples.
5. Explain **measurement scales of data** (Nominal, Ordinal, Interval, Ratio).
6. Explain **Sampling vs Population** and **Statistic vs Parameter**.
7. Explain **Random Variable**. Differentiate between **Discrete and Continuous**.
8. Explain **Probability Distribution of a Continuous Random Variable** with diagram.
9. Explain **Normal Distribution** with properties and diagram.
10. Explain **Binomial Distribution** with example (coin toss).
11. Explain **Conditional Probability** with equation and example.
12. Explain **Bayes Theorem** with example.
13. Explain **Types of Data Analysis**: Descriptive, Exploratory, Inferential, Predictive.
14. Explain **Misconceptions of Data Science**.
15. Explain **Data Science Life Cycle** with neat diagram.
16. Explain **Central Tendency vs Measures of Spread**.
17. Explain **Histogram, Box Plot, Scatter Plot** and their utility.
18. Explain **Simpson's Paradox** with example.
19. What is **Data Dredging**?
20. Explain **Pair Plot** and how to interpret it.
21. Explain **Data Pre-processing** – importance and phases.
22. Differentiate between **Data Cleaning, Data Curation and Data Pre-processing**.
23. Explain **Data Cleaning methods and best practices**.
24. Explain **Data Streams** and their characteristics.
25. **Data Streams vs Databases (DSMS vs DBMS)** with diagram.
26. Why is **Data Stream Mining** challenging?
27. Explain **Ad-hoc Queries vs Standing Queries** in data streams.
28. Explain **Landmark Model vs Sliding Window Model**.
29. Explain **Bloom Filtering** with example.
30. What is **Big Data**? Explain its **characteristics (V's)**.
31. Differentiate between **Big Data and Data Warehouse**.
32. Explain **Distributed File System** features in Big Data.
33. Explain **HDFS architecture** (Master–Slave).
34. Why is **HDFS** used for **Big Data processing**?
35. Explain **Hadoop-1 vs Hadoop-2 (YARN)**.
36. Explain **MapReduce paradigm** with **Word Count example**.
37. Explain **Map, Shuffle, Sort and Reduce** operations.
38. Explain **Rank computation** using **MapReduce**.
39. Explain **Apache Spark architecture and features**.
40. Differentiate between **Apache Spark and Hadoop**.
41. Explain **Spark Ecosystem components**: Spark Core, SQL, Streaming, MLlib, GraphX.
42. Explain **Hive architecture and utilities**.
43. What are **NoSQL databases**? Why are they used?
44. Compare **NoSQL vs RDBMS**.
45. Explain **Key-Value NoSQL databases** and use cases.
46. Explain **Column-based databases** and features.
47. Explain **Graph-based NoSQL databases** and when to use them.
48. Explain **HBase** and its applications.
49. Explain **Link Analysis** and its purpose in Data Science.
50. Explain **PageRank algorithm** with example.
51. Explain **Spider Trap and Dead End problem** in PageRank and solutions.
52. Explain **Sensitive PageRank**.
53. Explain **Similarity between two documents** with example.
54. Explain **Jaccard Similarity** with example.
55. Explain **Shingling technique**.
56. Compare **Minhashing vs Locality Sensitive Hashing (LSH)**.
57. Explain **Cosine Distance vs Edit Distance**.
58. Compare **Euclidean Distance vs Cosine Distance**.
59. Explain **Recommendation System** concept.
60. Explain **Content-based Recommendation System**.
61. Explain **Collaborative Filtering**.
62. Explain **Decision Tree, Entropy and Information Gain**.
63. Compare **Decision Tree vs Random Forest**.
64. Explain **Partitioning vs Pruning**.
65. Explain **Clustering types**.
66. Explain **K-Means Clustering** steps.
67. Explain **Association Rules**.
68. Explain **Predictive Analysis**.
69. Explain **Time Series Analysis**.
70. Short note on **SVM**.
71. Explain **Environment of R and Global Environment**.
72. Explain **Vectors, Lists, Matrices, Data Frames** in R.
73. Explain **Factors in R** and their characteristics.
74. Explain **String handling in R** with example.
75. Explain **Data Frames in R** and their features.
76. R program to **create and multiply two matrices**.
77. R commands for **Bar chart, Box plot, Histogram, Line graph, Scatter plot**.
78. Explain **Linear Regression** and its implementation in R.
79. Explain **Logistic Regression** and R function used.
80. Steps to perform in R: Logistic Regression, K-Means, Random Forest, Association Rules, Time Series.
81. JSON file handling in R and conversion to Data Frame.
82. Difference between **Descriptive vs Inferential analysis** with example
83. **Outliers** – effect on mean vs median
84. **ETL process** in data preprocessing
85. **Real-time stream processing examples**
86. **Why HDFS is not suitable for small files**
87. **RDD concept in Spark** (basic)
88. **Accuracy vs Precision vs Recall** (short note)
89. **Overfitting vs Underfitting**
90. **Supervised vs Unsupervised learning**
91. **Why normalization is needed in distance measures**